Research Notes on Expectation maximization algorithm

Expectation maximization algorithm

>> Section 1

L (θ ; X) = p (X ■ θ) = ∫ p (X , Z ■ θ) d Z = ∫ p (X ■ Z , θ) p (Z ■ θ) d Z {\displaystyle {\begin{aligned}L({\boldsymbol {\theta }};\mathbf {X})=p(\mathbf {X} \mid {\boldsymbol {\theta }})&=\int p(\mathbf {X} ,\mathbf {Z} \mid {\boldsymbol {\theta }})\,d\mathbf {Z} \\&=\int p(\mathbf {X} \mid \mathbf {Z} ,{\boldsymbol {\theta }})p(\mathbf {Z} \mid {\boldsymbol {\theta }})\,d\mathbf {Z} \end{aligned}}}

>> Section 2

Let x = (x 1 , x 2 , … , x n) {\displaystyle \mathbf {x} =(\mathbf {x} _{1},\mathbf {x} _{2},\ldots ,\mathbf {x} _{n})} be a sample of n {\displaystyle n} independent observations from a mixture of two multivariate normal distributions of dimension d {\displaystyle d} , and let z = (z 1 , z 2 , … , z n) {\displaystyle \mathbf {z} =(z_{1},z_{2},\ldots ,z_{n})} be the latent variables that determine the component from which the observation originates.

>> Section 3

where x ^ k {\displaystyle {\widehat {x}}_{k}} and x ^ k + 1 {\displaystyle {\widehat {x}}_{k+1}} are scalar state estimates calculated by a filter or a smoother. The updated model coefficient estimate is obtained via

>> Section 4

log ■ L (θ ; x , z) = ∑ i = 1 n ∑ j = 1 2 I (z i = j) [log ■ τ j − 1 2 log ■ | Σ j | − 1 2 (x i − μ j) ■ Σ j − 1 (x i − μ j) − d 2 log ■ (2 π)] , {\displaystyle \log L(\theta ;\mathbf {x} ,\mathbf {z})=\sum _{i=1}^{n}\sum _{j=1}^{2}\mathbb {I} (z_{i}=j)\left[\log \tau _{j}-{\tfrac {1}{2}}\log |\Sigma _{j}|-{\tfrac {1}{2}}(\mathbf {x} _{i}-{\boldsymbol {\mu }}_{j})^{\top }\Sigma _{j}^{-1}(\mathbf {x} _{i}-{\boldsymbol {\mu }}_{j})-{\tfrac {d}{2}}\log(2\pi)\right],}

>> Section 5

Relation to variational Bayes methods EM is a partially non-Bayesian, maximum likelihood method. Its final result gives a probability distribution over the latent variables (in the Bayesian style) together with a point estimate for θ (either a maximum likelihood estimate or a posterior mode). A fully Bayesian version of this may be wanted, giving a probability distribution over θ and the latent variables. The Bayesian approach to inference is simply to treat θ as another latent variable. In this paradigm, the distinction between the E and M steps disappears. If using the factorized Q approximation as described above (variational Bayes), solving can iterate over each latent variable (now including θ) and optimize them one at a time. Now, k steps per iteration are needed, where k is the number of latent variables. For graphical models this is easy to do as each variable's new Q depends only on its Markov blanket, so local message passing can be used for efficient inference.

>> Section 6

μ 2 (t + 1) = ∑ i = 1 n T 2 , i (t) x i ∑ i = 1 n T 2 , i (t) , Σ 2 (t + 1) = ∑ i = 1 n T 2 , i (t) (x i − μ 2 (t + 1)) (x i − μ 2 (t + 1)) ■ ∑ i = 1 n T 2 , i (t) . {\displaystyle {\begin{aligned}{\boldsymbol {\mu }}_{2}^{(t+1)}&={\frac {\sum \limits _{i=1}^{n}T_{2,i}^{(t)}\mathbf {x} _{i}}{\sum \limits _{i=1}^{n}T_{2,i}^{(t)}}},\\[1ex]\Sigma _{2}^{(t+1)}&={\frac {\sum \limits _{i=1}^{n}T_{2,i}^{(t)}\left(\mathbf {x} _{i}-{\boldsymbol {\mu }}_{2}^{(t+1)}\right)\left(\mathbf {x} _{i}-{\boldsymbol {\mu }}_{2}^{(t+1)}\right)^{\top }}{\sum \limits _{i=1}^{n}T_{2,i}^{(t)}}}.\end{aligned}}}

>> Section 7

Q (θ ■ θ (t)) {\displaystyle Q(\theta \mid \theta ^{(t)})} being quadratic in form means that determining the maximizing values of θ {\displaystyle \theta } is relatively straightforward. Also, τ {\displaystyle \tau } , (μ 1 , Σ 1) {\displaystyle ({\boldsymbol {\mu }}_{1},\Sigma _{1})} and (μ 2 , Σ 2) {\displaystyle ({\boldsymbol {\mu }}_{2},\Sigma _{2})} may all be maximized independently since they all appear in separate linear terms. To begin, consider τ {\displaystyle \tau } , which has the constraint τ 1 + τ 2 = 1 {\displaystyle \tau _{1}+\tau _{2}=1} :

>> Section 8

Alternatives EM typically converges to a local optimum, not necessarily the global optimum, with no bound on the convergence rate in general. It is possible that it can be arbitrarily poor in high dimensions and there can be an exponential number of local optima. Hence, a need exists for alternative methods for guaranteed learning, especially in the high-dimensional setting. Alternatives to EM exist with better guarantees for consistency, which are termed moment-based approaches or the so-called spectral techniques. Moment-based approaches to learning the parameters of a probabilistic model enjoy guarantees such as global convergence under certain conditions unlike EM which is often plagued by the issue of getting stuck in local optima. Algorithms with guarantees for learning can be derived for a number of important models such as mixture models, HMMs etc. For these spectral methods, no spurious local optima occur, and the true parameters can be consistently estimated under some regularity conditions.

>> Section 9

σ ^ w 2 = 1 N ∑ k = 1 N (x ^ k + 1 − F ^ x ^ k) 2 , {\displaystyle {\widehat {\sigma }}_{w}^{2}={\frac {1}{N}}\sum _{k=1}^{N}{({\widehat {x}}_{k+1}-{\widehat {F}}{\widehat {x}}_{k})}^{2},}

>> Section 10

The expectation of log ■ L (θ ; x i , Z i) {\displaystyle \log L(\theta ;\mathbf {x} _{i},Z_{i})} inside the sum is taken with respect to the probability density function P (Z i ■ X i = x i ; θ (t)) {\displaystyle P(Z_{i}\mid X_{i}=\mathbf {x} _{i};\theta ^{(t)})} , which might be different for each x i {\displaystyle \mathbf {x} _{i}} of the training set. Everything in the E step is known before the step is taken except T j , i {\displaystyle T_{j,i}} , which is computed according to the equation at the beginning of the E step section. This full conditional expectation does not need to be calculated in one step, because τ and μ/Σ appear in separate linear terms and can thus be maximized independently.